

Topic Modeling Analysis of Jihadists' Dark Web Forums

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Abstract

Terrorist groups have migrated to encrypted and hidden platforms such as the Dark Web forums for communication. This article applies natural language processing models to uncover topics of three Dark Web forums of Jihadist groups. Some forum members sympathize with and support terrorist organizations. Religion is the most prevalent topic in all forums. Other topics include terrorism, terror attacks, support for the Mujahideen fighters, relationships, selling electronics, and fake identity cards. Implications for counterterrorism include topic modeling in real-time classification, removal of online terrorist content, and monitoring religious forums, as terrorist groups use religion to justify their goals and recruitment.

Keywords: cyberterrorism, Dark Web, topic models, topic networks, natural language processing

Introduction

Terrorist groups and their supporters use the Internet to further their ideologies and goals, posing national security and counterterrorism challenges. Terrorism is a form of communicative violence (Aly et al., 2017). As such, terrorist groups use the Internet, specifically social media platforms, messaging apps, and Dark Web forums, to spread propaganda, recruitment, radicalization, fundraising, and communication. Thus establishing their appeal and popularity with a low risk of being captured (Behr et al., 2013; Bloom et al., 2019; Conway, 2017; Malik, 2018; Sageman, 2011; Weimann, 2016a, 2016b). Social media companies and counterterrorism officials are removing terrorist accounts and websites from the Internet. As a result, the groups have moved to the Dark Web, the part of the Internet that is hidden, not indexed, unregulated, and therefore not accessible via search engines. The Dark Web provides anonymity, the likelihood of evading arrests, and the removal of terrorist content, unlike the surface web (Weimann, 2016b). Thus, beneficial to terrorist groups who migrated to the encrypted platforms to protect their information from being removed and further disseminate news and propaganda, fundraise, and recruit. Individuals seeking terrorist content can easily find it on the Dark Web without being monitored or detected by law enforcement (Malik, 2018). Additionally, forums on the Dark Web encourage

radicalization and a “self-starter” type of terrorism that enables individuals to commit violence on behalf of the terrorist group, a type of “crowdsourcing terrorism” (Malik, 2018).

Before the growth of the Internet, most terrorist networks resulted from face-to-face interactions among friends and family (Sageman, 2011). The Internet has allowed like-minded individuals to interact and form virtual communities. As such, active online discussions about ideologies with friends or strangers and the sense of belonging within the virtual community or the virtual Muslim community (Ummah) have the potential to change someone’s mind rather than just reading impersonal stories (Bowman-Grieve, 2009; Erez et al., 2011; Scrivens et al., 2018). Forum users post and share their views and knowledge, and offer support and justification for engaging in ideologically motivated violence. The virtual networks comprise unaffiliated sympathizers, propagandists, fighters, recruiters, and rivals. Forums, not passive websites, are crucial in radicalization and recruitment. Thus, creating opportunities for self-radicalization, connection, and interaction of like-minded people globally while facilitating the promotion, popularization, and operation of radical ideologies and ideologically motivated violence.

While existing research has highlighted important ways terrorists use the Internet, specifically social media. An exploration of terrorists' use of the Dark Web forums, the topics, and the network structure of the forums used by terrorist groups and their supporters has been limited. The current study addresses the gap by examining three Dark Web forums of terrorist groups and their supporters using natural language processing techniques, specifically latent Dirichlet allocation (LDA) topic models (Blei, 2012; Blei et al., 2003) and topic network analysis (Walter & Ophir, 2019). Topic models are generative probabilistic models used for text mining and uncovering latent topics from documents. In analyzing terrorists’ online platforms, topic models are significant in automatically detecting large-scale terroristic content from the Internet. The paper is organized as follows: The first section describes the technological structure of the Internet, specifically the Dark Web. Followed by a discussion of related work in Dark Web forums and terrorist groups. The research design, methods, data, and results are explained in the following sections. Finally, the study concludes by discussing the Dark Web forums and their network structures, implications of the results, and limitations.

The Dark Web

The Internet is divided into three layers: surface, deep, and dark. Most people use the surface web or the Internet, which is reachable, indexed, and accessed via browsers such as Google, Bing, and Yahoo. Information from the surface web is visible to all who want access, and there are no restrictions. The Deep Web is the part of the Internet that is hidden, not indexed, and therefore not accessible via search engines. Access to Deep Web content requires sign-in passwords and is behind paywalls. For instance, medical records, membership websites, social media platforms, and online banking sites are part of the Deep Web. The Deep Web is approximately 400 to 500 times larger than the surface web, holding 400 times more content and 7,500 terabytes of information than 19 terabytes on the surface web (Chertoff, 2017; Malik, 2018). The Dark Web, or Darknet, hereby referred to as the Dark Web, exists within the Deep Web. The Dark Web is even harder to access, is unregulated, and contains approximately 0.01% of the content as opposed to the Deep Web's estimated 90% of content hosted on the internet (Malik, 2018; Weimann, 2016b).

The Deep Web is accessed for anonymity, usually for legal purposes, and access is granted via passwords, encryption, or specific software. The Dark Web is a layer deeper into the Deep Web and is intentionally hidden, accessed for anonymity, and sometimes illegal activities. Access to the Dark Web is granted through unique software programs supporting encrypted channels. Connections and networks can only be accessed by specific software, configurations, or authorizations such as the Tor (The Onion Router) or I2P (Invisible Internet Project). Dark Web sites provide anonymity and prevent monitoring of Internet users through encryption. For instance, the commonly used Dark Web browser, the Tor browser (The Onion Router), uses multilayered encryption that relays traffic and encrypts it three times. It is also used to block trackers and ads and defend against surveillance and fingerprinting. Onion routing routes traffic through multiple servers and encrypts to its destination (Tor Project, 2022). Based on multi-layer encryption and routing, users access the Dark Web sites securely, without revealing their location or identity, and anonymize visited websites (Chertoff, 2017; Grabosky, 2016; Malik, 2018). Ensuring the identity of users and their computers are anonymous in case of surveillance.

The Dark Web is accessed and used for both legitimate and illegitimate reasons. Legitimate users of the Dark Web include journalists, whistleblowers, activists, web users seeking to protect their privacy, or users who want to access hidden sites and services that may have been restricted or censored (Chertoff, 2017; Finklea, 2017; Mirea et al., 2019; Monk et al., 2018; Weimann, 2016b). Aside from the anonymity

users benefit from while on the Dark Web, the site is useful for cybercriminals. It should be noted that the Dark Web does not necessarily increase criminal activities; it is used as a tool to carry out illicit activities (Mirea et al., 2019), such as selling and buying illegal goods. Prior to their removal, marketplaces such as the Silk Road, AlphaBay, and DreamMarket were used for the advertisement and trading of illicit goods, including drugs, weapons, stolen identities, credit card details, child pornography, hackers, and hacking software for hire (Finklea, 2017; Gupta et al., 2021; Mirea et al., 2019). While law enforcement has shut down the marketplaces, new versions are recreated or moved to forums within the Dark Web where criminals communicate and perform their illegal transactions (Berton, 2015; Gupta et al., 2021), posing challenges for the detection and swift removal of criminals on the Dark Web.

Terror on the Dark Web Forums

As the terrorist platforms and materials on the surface web are monitored and removed, the groups migrate to the Dark Web, where anonymity is provided if not guaranteed. Private virtual spaces are encrypted. Administrators must verify members; membership is by referral only, and links and instructions for accessing the Dark Web forums are shared on social media accounts or other encrypted platforms (Berton, 2015; Malik, 2018). The Dark Web sites provide a 'safe space' for the groups and their supporters to interact, communicate, and access the information without being monitored. Additionally, media jihad is the production, distribution, promotion, collection, and redistribution of jihadist content as an obligation to incite violence against enemies (Erez et al., 2011). As such, the 'media mujahideen' (Malik, 2018) amplifies terrorists' online messaging campaign by sharing and reposting the official content of the groups across social media platforms and Dark Web forums. At times, terrorist groups compete for control of Dark Web forums and their supporters. For example, *Shumukh al-Islam* is a forum alternating between ISIL and al-Qaeda supporters (Berton, 2015).

Terrorists and their supporters use the Dark Web forums to discuss religion and jihadist ideologies (Abbasi & Chen, 2007; J. R. Scanlon & Gerber, 2015). Forums are the center of online jihadist activism; terrorists control them, and members can ask questions or contribute to discussion topics and private communications (Erez et al., 2011). The jihadist discourse online focuses on religious and ideological indoctrination, framed by references to the Quran and the early history of Islam (Erez et al., 2011). One can find texts such as the 'Terrorists' Handbook' and 'Explosives Guide' that can be purchased via the Dark Web marketplaces specializing in selling weapons (Berton, 2015). However, with the inaccessible nature of the Dark Web and with most people

only accessing the surface web, mass recruitment rarely takes place on the Dark Web. Instead, the groups tend to draw interested supporters from the surface web to the more secure Dark Web platform for further interaction, indoctrination, planning, and launching terror attacks (Malik, 2018).

The Dark Web forums encourage radicalization (Malik, 2018). The more time individuals spend online interacting with terrorists and their supporters, the more likely they may be radicalized into violent or radical social networks. A user can be defined as radicalized based on the content they post or share, i.e., anti-Western, pro-terrorist rhetoric, if they post very negative and inciting comments to the forum. For example, a long-time discussion forum member who posts harmful material over time suggests a dedication to an extremist movement. Or an author who participates for a short time but posts very radical views implies the user has become radicalized (Scrivens et al., 2018, 2021). However, it is difficult to classify users and their online posts as an indicator of radicalization (Ahmad et al., 2019; J. R. Scanlon & Gerber, 2015; Scrivens et al., 2018, 2021). Researchers have found that classifying a text or user as extremist is challenging due to the different kinds of extremism and ideologies, various targets, and different ways of expression and language. However, a pro-terrorist material and discussion forum would include terms and topics such as a call for jihad, Islam, mujahideen (members of the Taliban), apostate, caliphate, Ummah (Muslim community with jihadists worldview, nation), crusaders, promote violence towards the West, Christians, Jews, call for martyrdom, suicide operations, infidels (enemies, or non-believers), and offensive or abusive language such as kuffar (non-believers, non-Muslims). Additionally, Jihadist forum posts can be classified into themes based on shared information, such as religious preaching, instruction or training, and social interactions (Erez et al., 2011; Rowe & Saif, 2016).

The groups' online content and strategy are conducive to radicalization, as users seeking justification for joining or intending to join terrorist groups are likely to be indoctrinated by these messages. Additionally, terrorists such as ISIS online recruitment process follows a specific trajectory: (i) discovery of a potential recruit, or a recruit discovers ISIS; (ii) supporters form a community around potential recruits; (iii) potential recruits are encouraged to isolate and cut ties from their social bonds (families, friends), (iv) move to private or encrypted messaging spaces for conversations, and lastly, identify and encourage action to be taken by the recruit, whether to travel and join IS or carry out an attack at home (Berger, 2016; Berger & Strathearn, 2013). While the content attracts sympathizers and would-be extremists to frequent and participate

within the platforms, recruiters monitor these virtual communities to find potential recruits.

Once in the virtual community, users' language and social interactions change over time, mirroring those within their online community. As time passes, the users in these platforms have roles, including producing and aggregating content, translating and curating content across platforms, or passively consuming the information (Bloom et al., 2019; Shehabat & Mitew, 2018). As participants feel comfortable and welcomed on these platforms, skeptics and lurkers will be encouraged to share their beliefs and become more vocal. In addition, since participation in these forums is an active process, users tend to select platforms most compatible with their views (Piazza & Guler, 2019; Sageman, 2011). The Dark Web is an ideal ecosystem for terrorist groups; compared to the surface web, they can avoid censorship, discuss and communicate with their supporters, protect their identity and supporters, store their content without removal, and fundraise anonymously.

Methods

This analysis applies topic modeling algorithms to discover themes and topics automatically based on the co-occurrence and frequency distributions of words (Blei, 2012). While the topic structure is hidden, the documents and words are observed (Blei, 2012). The high-probability words in each topic distribution can be easily interpreted as a topic (Baumer et al., 2017). The unsupervised latent Dirichlet allocation (LDA) topic model does not require prior annotations or data labeling (Blei, 2012; Blei et al., 2003). Therefore, there is no need to manually label a data sample before identifying the topics. The researcher must only specify the optimal topic numbers and label the output. LDA models assume a topic is a distribution over vocabulary, i.e., a set of words, and a document is a mixture of topics. Topics can be inductively observed through topic modeling, network analysis, and community detection (Walter & Ophir, 2019). Community detection identifies groups of nodes, specifically, topics that are more densely connected than the rest of the network.

Data

Data used for analysis was accessed from the University of Arizona Artificial Intelligence Lab's Dark Web Forums (AZSecure-data.org). The forums were collected during the lab's study of the international Jihadi social media and movement (Chen, 2011; Chen et al., 2008). Three forums (see Table 1) were analyzed; the first is *Ansar1*, an English-language forum with 29,492 posts and 11,244 threads made by 382 members between

12/8/2008 and 1/20/2010. The second forum *Gawaher* is an English-language Islamic forum dedicated to discussing the Islamic world and Islam issues. Some of the forum members sympathize with radical Islamic groups. This forum entails 372,499 posts and 53,235 threads made by 9,269 members between 10/24/2004 and 6/7/2012. Finally, the *Islamic Network* forum is dedicated to various topics of interest to Muslims, ranging from theology and world events. Some members sympathize with and support terrorist organizations. This forum has 91,874 posts and 13,995 threads created by 2,082 members between 6/9/2004 and 11/10/2010.

Table 1. Dark Web Forum Data

Forums	Years	# Posts	# Threads	# Users
Ansar1	2008 - 2010	29,492	11,244	382
Gawaher	2004 - 2012	372,499	53,235	9,269
Islamic Network	2004 - 2010	91,874	13,995	2,082

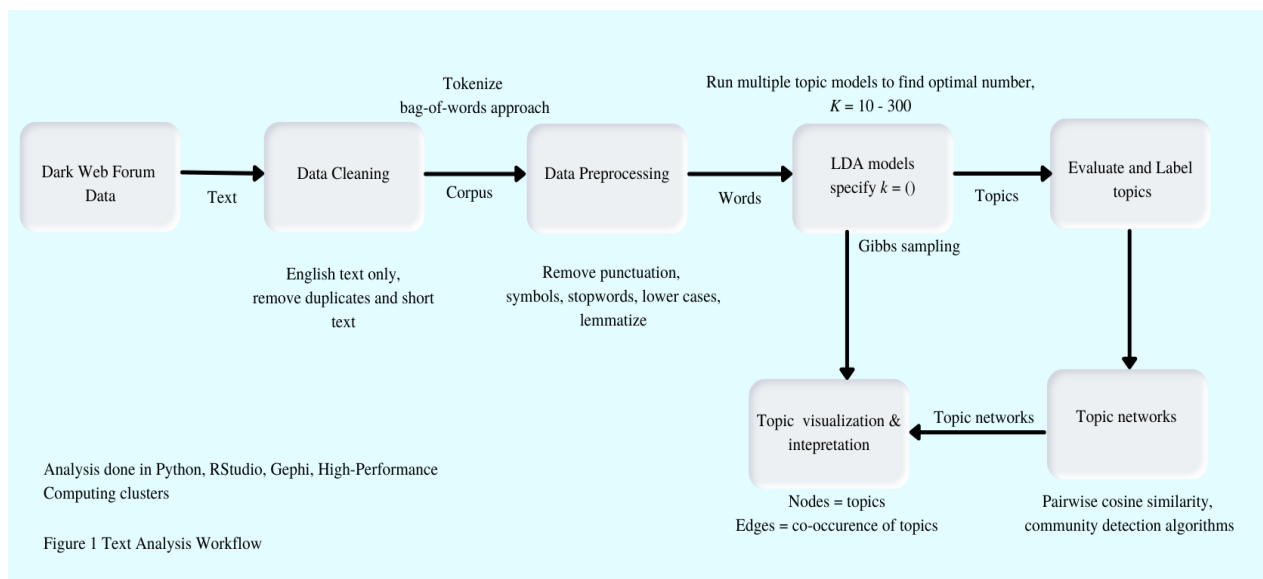


Figure 1. Text analysis workflow

Text Analysis

Data Pre-processing

Text analysis (see Figure 1) was conducted using R and Python statistical tools and LDA packages. In LDA, the text collection to be analyzed is called corpus (body of text), for instance, the *Ansar* Dark Web forum. An item within the corpus is a document, i.e., the message/post, and words within the document are called terms. One must first process text into a structured, machine-readable format to analyze it. While text analysis tools include translation packages that analyze multilingual text. Only English content were included in the analysis. Short texts and duplicates were also removed from the sample. The next step was to tokenize the text and break down the texts/sentences into tokens using the bag-of-words approach. The approach analyzes groups of words together, representing words in a phrase disregarding the order in which they appear (Lucas et al., 2015; Silge & Robinson, 2017), resulting in a matrix with unique terms and word frequencies.

Punctuations, numbers, and symbols were removed, and capital letters were changed to lowercase. Common stop-words such as "the," "and" "a" were also removed as they are not helpful for the detection of relevant topics. Additionally, words that appeared frequently and infrequently were filtered and removed. Similar to Walter & Ophir (2019) and Maier et al. (2018), words that occurred in more than 99% of the documents or less than .5% were removed. This process reduces the size of the corpus vocabulary, enhancing the algorithm's performance (Lucas et al., 2015; Maier et al., 2018). Stemming and lemmatization are NLP processes used to reduce inflection forms of a word to a common base or dictionary form, i.e., its stem or root, lemma. However, due to the outputs of the stemming process often not making sense, it is difficult to interpret the words, thus impacting the topic quality (Maier et al., 2018; Walter & Ophir, 2019) by degrading the topic stability (Schofield & Mimno, 2016). The final preprocessing step was to lemmatize some of the words to their lemma forms, i.e., 'study,' 'studies,' and 'studying,' are collapsed to 'study.'

LDA Models

The next step is to train the LDA models using Gibbs sampling iteratively. This algorithm uses thousands of iterations through the documents, considers the topic proportion of each word token, and places words that frequently occur together in a document into the same topic (Baumer et al., 2017). The number of topics, K , is a

parameter that must be defined for an LDA model. The number of topics can range from 2 to 100 based on the research questions, the overall size of the corpus, and evaluation metrics. The more topics, the more specific the topic outputs will be. On the other hand, too many topics may lead to a similar outcome that is not meaningful, or too few topics may lead to broad outputs that combine different terms that should have been separated (Maier et al., 2018).

To assess and evaluate the ideal number of topics for each Dark Web forum, a range of topic numbers from 10 - 300 were estimated and evaluated using four different model fit indicators from the *LDAtuning* package. The model fit metrics are based on the accuracy (Arun et al., 2010), density (Cao et al., 2009), latent concept modeling (Deveaud et al., 2014), and Markov chain Monte Carlo algorithm (Griffiths & Steyvers, 2004). The results from the model fit metrics were used to select a final number of topics for each dataset. The next step was to evaluate the relevant topic numbers and rerun the model. Then, extract the theta and beta statistics from the output and plot, interpret, and label the topics. Similar to Walter & Ophir (2019), the results of the top words in the topics, words that occur frequently and are exclusive to each topic, and the original documents representing each topic were analyzed. The topics were then interpreted and labeled by examining the words of each topic and the documents. Only some topics in topic models are relevant or interpretable; some topics contained words that needed further clarification and were, therefore, challenging to interpret. After labeling the topics, a topic network was created using pairwise cosine similarity and community detection algorithms (Walter & Ophir, 2019). Nodes represent topics, and the edges are the co-occurrence of topics in the documents. Communities or clusters based on language co-occurrence were detected using the walktrap clustering algorithm (Pons & Latapy, 2005).

Results

Ansar1 Dark Web Forum

A topic is outlined by its probability distribution over words. The results include representative texts. Below, I highlight the ten most probable words for a given topic and the full representative text of these topics. For instance, topic 23, which I labeled 'Mujahideen Media,' has the following top 10 terms: *Mujahideen, terrorist, kill, report, destroy, Islam, Afghanistan, Allah, Emir, and puppet.*

The first complete text example was:

"[1] Mujahideen of Islamic Emirate of Afghanistan military operations against the kafirs, munafiqs, and the worshippers of Idols. This page is updated throughout the day as new operations are reported [...] the Mujahideen operations against the enemies of Islam terrorists in Afghanistan are reported to Theunjustmedia.com by the official Mujahideen of Islamic Emirate of Afghanistan spokesmen Qari Muhammad Yousuf and Zabeehullah Mujahid by e-mails." (Ansar1 Dark Web Forum).

Further interpretation of the representative texts suggests that the forum members posted and discussed news from the Mujahideen media. Another example is topic 29, labeled 'Advice for Jihadis,' which has the following top 10 words: *God, people, Muslim, religion, said, fight, one, victory, great, believe, faith, infidel.* This topic encourages supporters to join the fight for jihad, the Islamic religion, and their nation to prevent the West or non-Muslims from taking over their faith and country.

Gawaher Dark Web Forum

Gawaher forum has 67 topics and two communities (Figure 3); the largest community in green topics is primarily about religion, Islamic scripture, the Bible vs. Quran, converting to Islam, and non-Muslims. There were some philosophical discussions about the existence of God and evolution, and relationships. Topics within the smaller community in red were about the U.S. military and the U.S. in Iraq, terrorists, Israel, Palestine, Hamas, the Taliban, and the imprisonment and torture of terrorists by the U.S. A few of the topics not included in the network mainly were posts from the forum admin reminding the members about posting rules and that posting links was forbidden until one had submitted 50 posts.

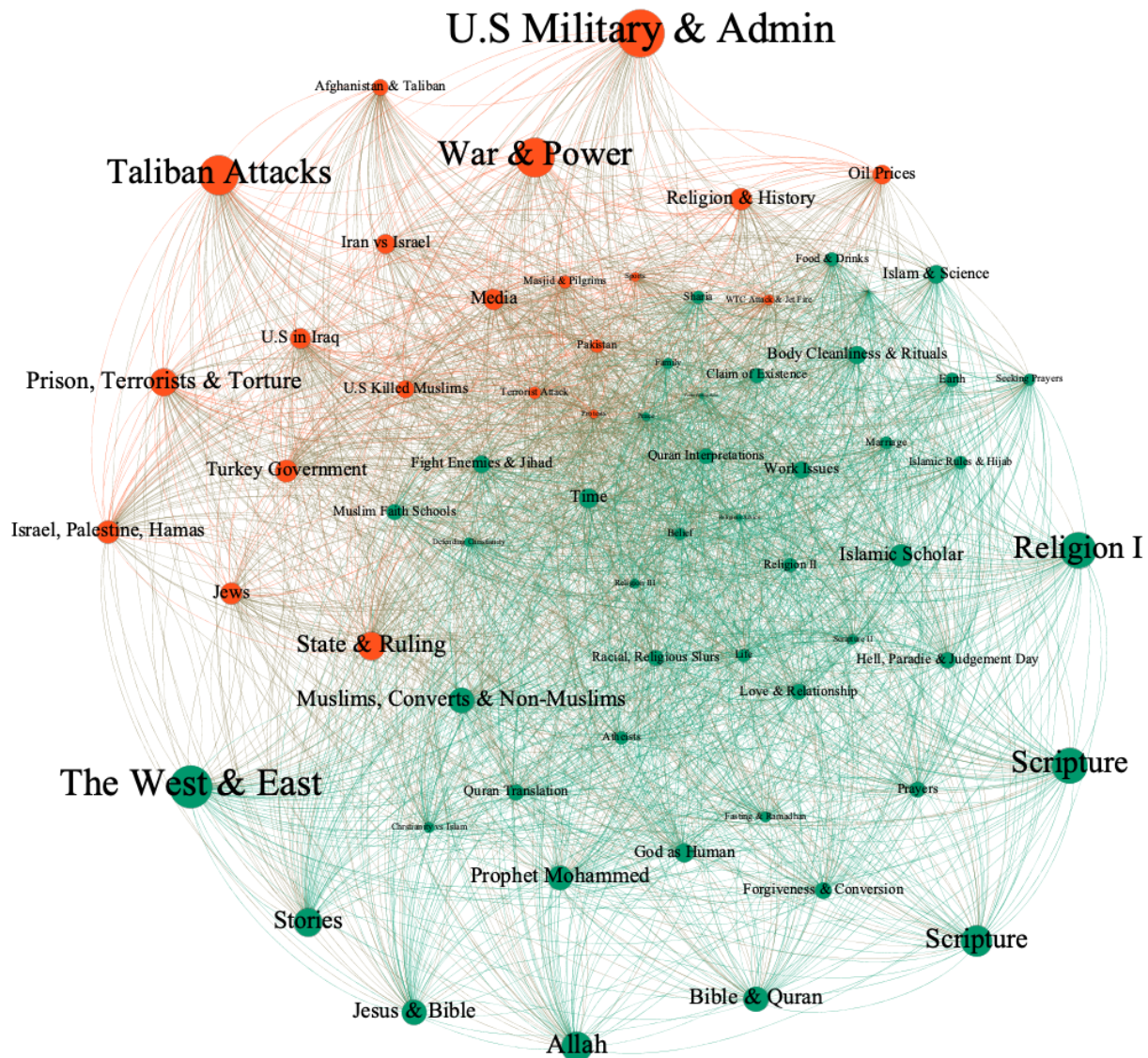


Figure 3 Gawaher Dark Web Forum Topic Network. The topic network has 67 Nodes representing topics, and 2211 Edges represent the co-occurrence of topics within the documents. The node size means the importance of each topic in the network. Color represents the community membership, weighted and undirected network.

The most prevalent topic in the Gawaher topic network is topic 80, labeled “U.S Military and Admin.” The top ten words include *official, new, president, unit, report, American, Washington, military, said, intelligence*. In topic 87, ‘Taliban Attacks,’ the top

words include: *said, kill, soldier, force, report, police, attack, bomb, military, Baghdad*. Another interesting topic was Topic 9, labeled “Forum Posting Rules”, the top words include: *forum, discussion, video, topic, can, site, read, please, may, website, thread, and member*. This topic shows how the forums are organized and how members use descriptive titles to attract volume and interactions.

Islamic Network Dark Web Forum

The *Islamic Network* forum resulted in 72 labeled topics and 3 communities (Figure 4). The 72 topics for the *Islamic Network* contributed to three major topic clusters. The topics in green discussed Guantanamo Bay and how prisoners are tortured and interrogated in that prison. Other topics were terrorism, US troops in Iraq, the death of soldiers from terrorist groups, the Taliban in Pakistan, and the relationship between Israel and Palestine. The topics in red/orange were a mix of electronics sold on the Dark Web, jihad, religion, the discussion on nationalism, people sharing stories, advertisements about the London talk by an Islamic scholar, investments, and money, advertisements about selling fake documents, and IDs. Finally, the cluster in blue had topics on religion, Quran recitations and verses, Allah and God, Imam, and Islamic schools/scholars.

In this forum, the most prevalent topic 14 is labeled ‘Guantanamo,’ top words include *prison, Guantanamo, torture, detained, court, case, release, charge, terror, British*. Another prevalent topic is topic 49, labeled ‘Electronics Ads,’ the most probable words for this topic include: *soni, nokia, apple, htc, Samsung, Ericson, phone, notebook, pioneer, pc, tv, ipod, new, digit*. Although not prevalent, topic 64, labeled ‘Fake Documents Ad,’ is interesting. The top words include *passport, fake, false, sell, new, id, buy, identity, driver, sale, and private*. This post was about selling fake documents, including passports, driver’s licenses, and other products for most countries. The topics were primarily about religion and few on relationships, terror attacks, and electronics sales in three clusters. Most interesting is the illegal activities conducted on the Dark Web forums, including forums used by terrorist groups and their supporters.

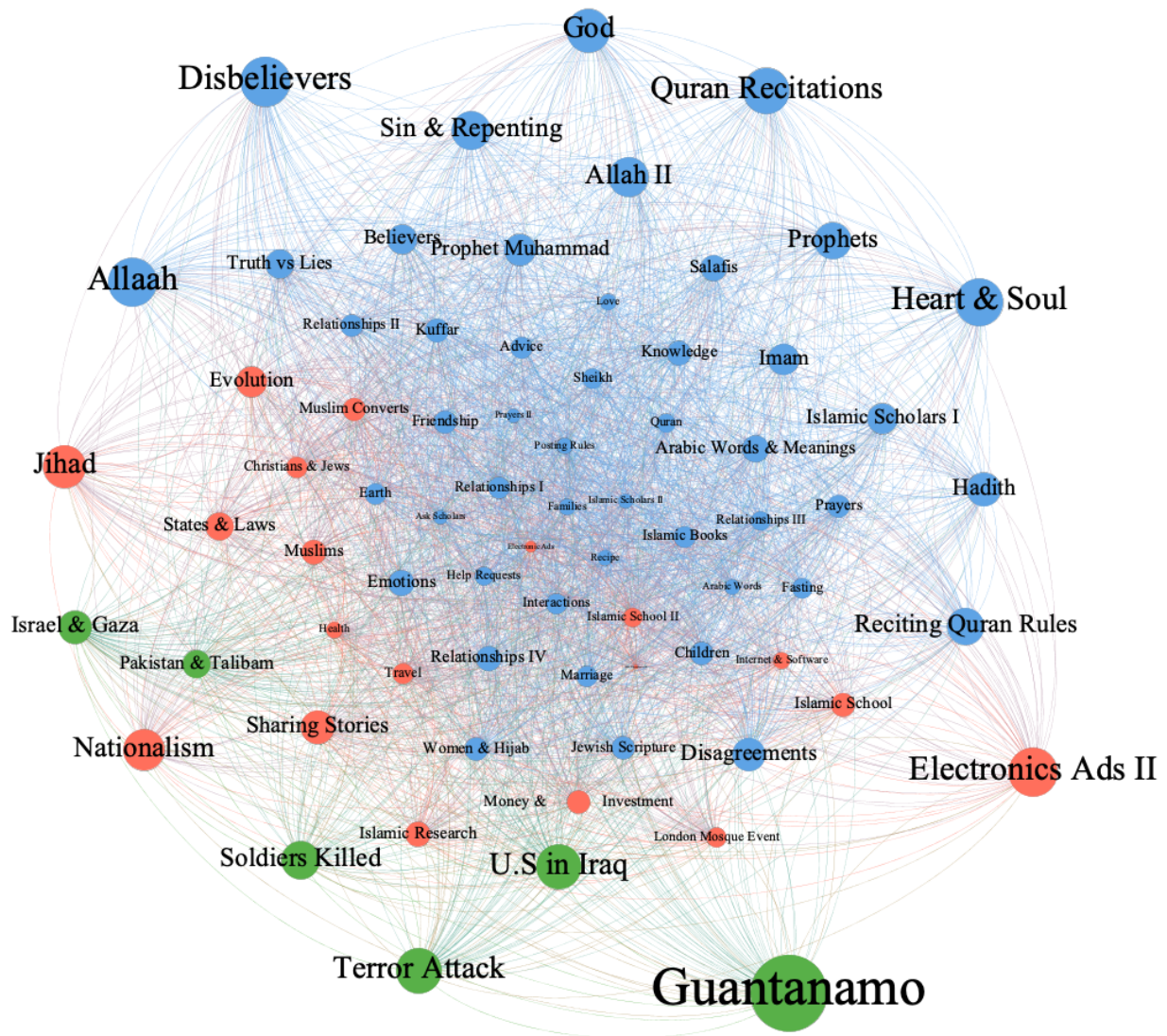


Figure 4 Islamic Network Dark Web Forum Topic Network. Topic network with 72 Nodes representing topics, 2556 Edges representing the co-occurrence of topics within the documents. The node size means the importance of each topic in the network. Color represents the community membership. The network is weighted and undirected.

Discussions and Conclusions

Terrorist groups and their supporters use the Internet to further their ideologies and goals, posing national security and counterterrorism challenges. To evade detection or

disruption, the groups and their supporters have moved to the Dark Web (Malik, 2018; Weimann, 2016b). LDA topic models identified coherent topics and clusters of the groups' Dark Web forums. The most prevalent topic in all three forums was religion. Topics about religion included Islam, Christianity, and Judaism, discussion about Allah and God, believers and disbelievers, Quran recitations, religious rituals, and philosophical discussions about religion and science. These topics are expected, as the forums were dedicated to the Islamic faith and issues concerning Muslims.

The topic of terrorism was prevalent in all forums, mainly within the *Ansar1* forum. Here, the discussion was about terrorists and attacks, support for the Mujahideen fighters, jihad, Al-Qaeda, Taliban, the U.S. in Iraq and Afghanistan, and the arrests and torture of terrorists and foreign fighters. One could argue that the discussion or posts about terror groups were created by the terrorist groups and their media centers and supporters, also known as 'media mujahideen' (Malik, 2018). Most of the jihadist discourse online involves religious/ideological indoctrination framed by references to the Quran, Hadith, and the early history of Islam (Erez et al., 2011). A few discussions were on relationships and marriages, advice, seeking help, health, food, selling electronics, and identity cards. Topics on selling fake IDs and documents indicate the presence of criminals in terrorist networks and possibly a partnership between criminals and terrorists, also known as a crime-terror nexus.

By clustering the topics into communities, one can highlight the frequency of language use from the corpus used by each community, in this case, the extent of discussions based on terrorism and/or Jihadism. For instance, in the *Ansar1* forum, 51% of the corpus was associated with the community on the left, in blue (Figure 2). In contrast, 49% of language used in the forum corpus is likely to be used by the community on the right, in red. Similarly, for the *Gawaher* forum (Figure 3), the topics in the green community were most likely to be associated with 69% of the language in the corpus and 31% for the topics in the red community. In the *Islamic Network* forum (Figure 4), the topics in blue are most likely to be associated with 65.3% of the words in the corpus, 26.4% for the topics in red, and 8.33% for the topics in the green community.

In addition to identifying the topics and the prevalence of language use, one can also classify extremist content from the Dark Web. Results suggest that from the 82 topics identified in the *Ansar1* forums, there is an equal mixture of religious and terrorism content. Most of the *Gawaher* forum's 67 topics were about religion (61%) and terrorism content (31%). The *Islamic Network* forum's 72 topics resulted in three classifications. The prevalent topic is religion (65.3%), a mixture of religion, legal and

illegal activities (26.4%), and terrorism (8.33%). The *Ansar1* forum was mainly dedicated to discussing or sharing terrorist content as opposed to the other two forums, which were especially about Islam and Muslims. It should be noted that the *Ansar1* forum is the English forum of the Arabic *Ansar AlJihad* Network, which is an invitation-only Jihadist forum popular with Western Jihadists (J. Scanlon & Gerber, 2014). Additionally, as mentioned elsewhere, topics on religion can also be classified as Jihadist discourse, using religion to justify their ideology.

LDA topic modeling has several advantages over the manual annotation of text data, especially when analyzing larger corpora. When analyzing big data generated by Internet users, automatic computational techniques are more effective, faster, and reliable. Topic models quickly discover hidden topics, thus providing real-time text analysis. The topics are inductively uncovered and require no previous knowledge of the content from the large-scale text data. Additionally, topic models offer insights into extensive unstructured text data instead of traditional structured data such as surveys and interviews. The keywords discovered from the topic models can further be utilized to organize and search for content in other text data. For instance, the keywords derived from the three Dark Web forums can be used to analyze or search other Dark Web forums or websites for similar topics.

Compared to manual annotation, topic models require less time and money and reduce researcher biases. Topic modeling is data-driven and thus requires very minimal human supervision. The only work for the researcher is to input a topic number and interpret and label the output. Hence, topic models illustrate that textual data can also be analyzed quantitatively and mitigate researcher bias. Additionally, compared to other text classification algorithms of unlabeled data, such as k-means clustering analysis, where documents can be classified into only one cluster, topic models are mixture models where each document is assigned a probability of belonging to a topic; therefore, a document can belong in different topics.

However, some limitations abound when using topic modeling. First, topic modeling can be computing intensive depending on the corpus size to be analyzed. An analysis can run for two days or weeks, depending on the computer's power. Secondly, language is ambiguous; not all topics can be labeled and interpreted into a recognizable theme. For instance, despite considering the full text and context, some of the topics in the *Gawaher* and the *Islamic Network* forums were challenging to interpret and label, were not meaningful, and thus omitted from the analysis.

Future directions may analyze forum members' contributions to the topics and their sentiments. For instance, using structural topic modeling, author-topic models, sentiment analysis, or other algorithms to classify the posts and members as radical or non-radical, based on the posts, time spent on the forum, engagement with others, i.e., responding to all threads, etc. Another future study may analyze the social network structure, including the key members and communities, to highlight how the members interact in the Dark Web forums.

From the classification of jihadist Dark Web forums, topic modeling can be applied automatically to analyze large-scale text from online extremist content. Detecting extremist content online using topic models is significant for analyzing users' beliefs and countering and/or disrupting their online narrative. As Jihadists continue to use the Internet for communication and recruitment, topic models are valuable techniques that automatically detect and classify their online content for crime prevention and security.

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